

Functional genetic report · longevity

Epigenetic Clock and Biological Age

Sérgio F · Report date: 02-10-2026

What part of the speed of epigenetic clocks comes from genes, and what genetics says about the biological age of the organism (fragility). Only genes (chip + local imputation + scores). The telomeres are in the report "Telomeres and Cell Senescence".

Contents

• Summary	3
• How to read this report	4
• Polygenic scores	5
• Intrinsic clock (Horvath/IEAA)	6
• 2.a generation clocks GrimAge, PhenoAge and Hannum	7
• Epigenetic machinery and clonal hematopoiesis	8
• Fragility: biological age of the whole organism	9
• Mechanisms of ageing	10
• Practical suggestions	11
• Limitations	12
• Methods and sources	13

Summary

17

SNPs analyzed

4

in the chip

6

imputed ($DR2 \geq 0.8$)

7

SNP not found in the data

5

Polygenic scores

Epigenetic clocks estimate biological age from DNA methylation. The intrinsic clock speed (IEAA, Horvath) is ~40% hereditary; that of the 2.a generation clocks (GrimAge, PhenoAge), which better predict mortality, is mainly environmental: tobacco, adiposity, inflammation, sleep, exercise and diet. Therefore, genetics gives only the starting point.

In summary: typical to favourable genetic starting point. The intrinsic clock is at the 48th percentile and the GrimAge in the 7 (small variants); PhenoAge (78) and Hannum (68) are on the typical-high side, with low coverage. There are missing alleles that accelerate the clock in TERT (rs2736099 GG) and TRIM59/KPNA4. The frailty score (943 thousand variants) is at the 75th percentile. Without the protective allele of TCL1A against clonal hematopoiesis.

Favourable

- **Intrinsic clock at the 48th percentile** (IEAA, 18 of 20 loci) and GrimAge at the 7th percentile (only 4 of 6 loci; indicative).
- **TERT rs2736099 GG and TRIM59/KPNA4 rs11706810 TT**: without the alleles that accelerate the intrinsic clock.
- **TERT rs7705526 heterozygous** (intermediate effect on clonal hematopoiesis) and MTHFR 677 only heterozygous.

Unfavourable

- **PhenoAge in the 78 and Hannum percentiles in the 68** (typical, high side; low coverage of variants).
- **TCL1A rs2887399 GG**: without the protective allele against the expansion of blood clones (CHIP) and Y loss (as ~65% of the population).
- **Fragility at the 75th percentile** (943 thousand variants; typical, high side).
- **Several loci of GrimAge and PhenoAge not visible** (infrequent variables, imputed with insufficient quality).

Interpretation: genetics does not condemn or guarantee: most of the speed of clocks that predict mortality (GrimAge, PhenoAge) is modifiable. The intrinsic clock, the most genetic, is exactly on average.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 4 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

The scores of epigenetic clocks use only the leading variants of GWAS Catalog (McCartney 2021, Lu 2018) and do not include infrequent variants that the chip does not read well; the coverage of PhenoAge and Hannum is ~60%. The frailty score is of complete genome.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Acceleration of the intrinsic clock (IEAA; 20 loci)	C_IEAA	18 of 20 (90%)	p48	Typical / no judgement
GrimAge Acceleration (6 loci)	C_GRIMAGE	4 of 6 (67%)	p7	Favourable
PhenoAge acceleration (13 loci)	C_PHENOAGE	8 of 13 (62%)	p78	Typical / no judgement
Acceleration of Hannum clock (10 loci)	C_HANNUM	6 of 10 (60%)	p68	Typical / no judgement
Fragility, general factor (943 126 variants)	PGS005221	883 191 of 943 126 (94%)	p75	Typical / no judgement

Intrinsic clock (Horvath/IEAA)

The Horvath clock, adjusted for the composition of blood cells (IEAA), measures an intrinsic cellular aging. The strongest loci are in TERT, 6p22 (KIF13A/NHLRC1) and 3q25 (TRIM59/KPNA4).

Polygenic scores of this theme: Acceleration of the intrinsic clock (IEAA; 20 loci) (C_IEAA): the 48th percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
TERT	rs2736099	GG	imputation DR2=0.88	A	0	0.34	TERT: the allele that stretches the telomeres accelerates the intrinsic clock (stem cells divide more). (moderate evidence)	Favourable
KIF13A/ NHLRC1	rs10949481	AA	imputation DR2=0.99	A	2	0.97	6p22.3: the strongest locus of the intrinsic clock (IEAA). (strong evidence)	Typical / no judgement
PIPOX/ MYO18A	rs7211375		SNP not found in the data	C	—	—	17q11.2: intrinsic clock. (moderate evidence)	SNP not found in the data
TRIM59/ KPNA4	rs11706810	TT	imputation DR2=0.98	C	0	0.43	3q25.33: intrinsic clock (first locus discovered, Lu 2016). (moderate evidence)	Favourable
TPMT/ NHLRC1	rs73724422	TT	imputation DR2=1.00	T	2	0.97	TPMT/NHLRC1: intrinsic clock (methyl metabolism). (moderate evidence)	Typical / no judgement
SELP	rs6687517	TC	imputation DR2=0.98	C	1	0.48	P-selectin: intrinsic clock. (weak evidence)	Typical / no judgement

Favourable

- TERT rs2736099 GG
- TRIM59/KPNA4 rs11706810 TT
- IEAA score at the 48th percentile

Unfavourable

- SELP rs6687517 TC (typical)

Interpretation: TERT paradox: the alleles that stretch the telomeres accelerate the intrinsic clock (stem cells divide more). Sérgio F does not have this allele. KIF13A/NHLRC1 AA and TPMT TT are the ~94% genotypes of the population.

2.a generation clocks GrimAge, PhenoAge and Hannum

GrimAge (based on plasma proteins and tobacco), PhenoAge (based on clinical biomarkers) and Hannum: the clocks that best predict disease and mortality.

Polygenic scores of this theme: GrimAge acceleration (6 loci) (C_GRIMAGE): the 7th percentile **Favourable**; Acceleration of PhenoAge (13 loci) (C_PHENOAGE): the 78th percentile **Typical / no judgement**; Acceleration of the Hannum clock (10 loci) (C_HANNUM): the 68th percentile **Typical / no judgement**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
SIN3B/ F2RL3	rs2227341		SNP not found in the data	A	—	—	F2RL3 (susceptible methylation site used in GrimAge): GrimAge. (strong evidence)	SNP not found in the data
APRG1	rs78131239		SNP not found in the data	G	—	—	3p22: GrimAge.	SNP not found in the data
ZNF106	rs28364559		SNP not found in the data	T	—	—	ZNF106: GrimAge.	SNP not found in the data
CREB3L3	rs73538174		SNP not found in the data	G	—	—	CREB3L3 (lipids and triglycerides in the liver): PhenoAge. (strong evidence)	SNP not found in the data
CREB3L3	rs8103978		SNP not found in the data	T	—	—	CREB3L3: PhenoAge.	SNP not found in the data
F5	rs9332539		SNP not found in the data	A	—	—	F5: Hannum clock. (moderate evidence)	SNP not found in the data
BLNK/ DNTT	rs941997	AT	imputation DR2=0.97	A	1	0.79	BLNK/DNTT (lymphocytes B and T): Hannum clock (reflection of blood cells). (moderate evidence)	Typical / no judgement

Favourable

- GrimAge at the 7th percentile (4 of 6 loci)

Unfavourable

- PhenoAge p78 and Hannum p68 (high side, coverage 60%)
- F2RL3, APRG1, ZNF106, CREB3L3 and F5: not visible

Interpretation: These loci have small effects and many are infrequent; the scores are indicative. BLNK/DNTT AT: the Hannum clock also depends on the composition of immune cells.

Epigenetic machinery and clonal hematopoiesis

DNMT and TET write and erase methylation; mutations acquired in DNMT3A and TET2 create clones of blood cells (CHIP), which accelerate clocks and cardiovascular disease.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
TCL1A	rs2887399	GG	chip	T	0	0.21	TCL1A: T allele reduces the expansion of blood stem cell clones (CHIP with TET2, ASXL1) and the loss of Y chromosome. (strong evidence)	Typical / no judgement
TERT CHIP	rs7705526	CA	chip	A	1	0.35	TERT rs7705526 A: longer telomeres and more clonal hematopoiesis (CHIP). (moderate evidence)	Typical / no judgement
MTHFR C677T	rs1801133	GA	chip	A	1	0.37	MTHFR 677T: less 5-metilfolato for methylation of DNA (see methylation report). (moderate evidence)	Typical / no judgement
DNMT3A	rs2289195	GA	chip	—	—	—	DNMT3A (methyltransferase de novo; the most mutated gene in CHIP): common polymorphism without defined effect. (weak evidence)	Typical / no judgement

Favourable TERT rs770526 CA (intermediate)	Unfavourable TCL1A rs2887399 GG (without protection against CHIP)
---	--

Interpretation: Genetic predisposition to CHIP is typical. MTHFR 677 GA: a lower methylfolate production allele. DNMT3A rs2289195: polymorphism without defined effect.

Fragility: biological age of the whole organism

The fragility (loss of reserve of several systems) is the clinical expression of biological age. The general fragility factor (Foot 2025) summarizes 30 deficits in 943 thousand variants.

Polygenic scores of this theme: Fragility, general factor (943 126 variants) (PGS005221): the 75th percentile
Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
Favourable				Unfavourable				
• None.				• None.				

Interpretation: General score of frailty at the 75th percentile. Genetic fragility shares a lot with depression, chronic pain and adiposity.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Epigenetic Derivative	PhenoAge p78, Hannum p68	IEAA p48; GrimAge p7; TERT and TRIM59 without accelerator alleles	Typical to favourable.
Loss of Epigenetic Fidelity / Epigenetic Alterations	MTHFR 677 GA	—	Small effect.
Clonal hematopoiesis (CHIP)	TCL1A GG (without protection)	TERT rs7705526 heterozygous	Typical.
Somatic Mutation Accumulation	TCL1A GG	—	Typical.
Hayflick's Telomeres and Limit	—	TERT rs2736099 GG (without clock acceleration)	See telomeres report.
Immunosenescence	BLNK/DNTT AT (Hannum)	—	Neutral.
Inflammaging	—	Low genetic GrimAGE	Favourable.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Exercise	Aerobic training + regular force: the factor with more data to slow down GrimAge and DunedinPACE. <i>suggestion (not applied)</i>	PhenoAge and Hannum on the upper side
Omega-3 and vitamin D	Keeping fish oil (omega-3 3.2-4 g/day) and D3 10 000 IU, as in the plan: in DO-HEALTH, omega-3 slowed down several epigenetic clocks. <i>suggestion (not applied)</i>	2 clocks.the modifiable generation
Diet	Mediterranean pattern with a lot of fiber (the better, the better), legumes, red fruits and olive oil; little processed meat. <i>suggestion (not applied)</i>	2.a generation clocks
Sleep and stress	Regular 7-8 h sleep and recovery; melatonin as in the plan (0.3-1 mg). <i>suggestion (not applied)</i>	GrimAge/PhenoAge
Visceral Adiposity	Reduce visceral adiposity (retatrutide and training already underway) reduces inflammation that accelerates PhenoAge and GrimAge. <i>suggestion (not applied)</i>	PhenoAge p78
Follow	If you want to follow, a 3.a generation clock (DunedinPACE) or GrimAge once a year, always from the same laboratory. <i>suggestion (not applied)</i>	Modifiable clocks
What not to do	Without telomerase activators (accelerated the intrinsic clock and have an oncological risk). Curcumin, luteolin, apigenin and quercetin in pause while ALT is high; senolytic pulse only fisetin + quercetin with normal ALT. <i>suggestion (not applied)</i>	V2 plan rules

Limitations

- Clocks measure methylation, not genes: this report only estimates the hereditary part of clock speed.
- The scores of the clocks use few leading variants; several infrequent variants of GrimAge and PhenoAge were not read with sufficient quality.
- The percentiles compare with the reference population (n=633).
- Only genes: does not use methylation tests or blood analysis.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (Beagle 5.5, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population (n=633).
- Effect alleles of GWAS Catalog (local copy), with the verified sense (clockwise acceleration = unfavourable); excluding studies from populations very different from those of reference.
- Scores: simple scores of GWAS Catalog for IEAA (20 loci), GrimAge (6), PhenoAge (13) and Hannum (10); PGS005221 (general frailty factor, Foote 2025). Percentages vs. reference population (n=633) with exactly the same variants.
- Main references: Horvath 2013; Lu 2016 and 2018 (GWAS of the clocks, Nat Common); McCartney 2021 (Genome Biol); Lu 2019 (GrimAge); Levine 2018 (PhenoAge); Belsky 2022 (DunedinPACE); Bischoff-Ferrari 2025 (DO-HEALTH, Nat Aging); Weinstock 2023 (TCL1A and CHIP, Nature); Foote 2025 (fragility, Nat Genet).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).